

Andy Sin

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PROFESSIONAL SUMMARY

Computer Engineering freshman at Cal Poly San Luis Obispo with applied experience in robotics software, autonomous control systems, embedded hardware, and PCB design. Led an FRC team's full migration to WPILib command-based Java, implementing swerve drive, sensor fusion with AprilTag vision, and autonomous path planning. Seeking internships in robotics, embedded systems, controls, or hardware/software integration.

TECHNICAL SKILLS

Languages: Java, Python, GDScript, C++

Robotics & Software: WPILib Command-Based, Git/GitHub, Phoenix Tuner X, PathPlanner, Choreo, Limelight, AprilTags, Godot, KMK

Hardware & Design: KiCad, Onshape, Raspberry Pi 4, Seeed XIAO RP2040, PCB Design, Soldering, 3D Printing, Multimeter

ENGINEERING EXPERIENCE

Lead Programmer & Competition Strategist | Aug 2025 - Jun 2026

Programming & Outreach Member | Aug 2024 - Jun 2025

FIRST Robotics Competition Team 3598, SEStematic Eliminators | Sacramento, CA

- Led the team's first migration from LabVIEW to WPILib command-based Java, programming swerve drivetrain, turret/shooter, intake, indexer, climber, autonomous routines, and NetworkTables while coordinating with mechanical and electrical subteams.
- Developed teleop/autonomous turret auto-aim and dynamic shot calculations using robot velocity and target distance; fused drivetrain odometry with Limelight AprilTag data in SwerveDrivePoseEstimator and scaled vision measurement std. deviations by distance.
- Created and adapted autonomous paths with Phoenix Tuner X, PathPlanner, and Choreo; served as lead strategist at the team's first event and alliance captain during the Half Moon Bay playoffs.
- Facilitated 15+ hands-on STEM workshops for students ages 3-14, supporting teamwide outreach that reached 9,000+ youth over three years.

PROJECTS

SinnerPad - 4-Key Digital Art Macropad | Independent Project

- Designed and built a four-key macropad using KiCad, a Seeed XIAO RP2040, Cherry MX-compatible switches, four SK6812MINI-E RGB LEDs, and a two-piece 3D-printed enclosure in Onshape; completed as a self-imposed one-day first hardware build.
- Soldered and assembled all components and configured KMK firmware for undo, redo, copy, and paste shortcuts; verified all key inputs and RGB lighting.

Shiba Arcade Game & Custom Cabinet | Hack Club | 2025

- Placed among the top 30 of 8,000+ participants in an international teen game-development challenge, earning an all-expenses-paid trip to Japan.
- Developed a Godot/GDScript fixed-turret wave shooter with procedurally randomized gameplay, upgrades, and original art/audio; built, painted, wired, and soldered a Raspberry Pi 4 (8 GB) arcade cabinet with joystick and button controls for public play.

EDUCATION

California Polytechnic State University, San Luis Obispo | San Luis Obispo, CA

B.S. Computer Engineering | Expected June 2030

HONORS & CERTIFICATIONS

Individual Honors: Hack Club Shiba Award; AP Scholar with Distinction; College Board First-Generation Recognition Award | 2025

FIRST Team Awards: Impact Award (2025, 2026); Team Spirit Award (2025, 2026); Judges' Award (2025)